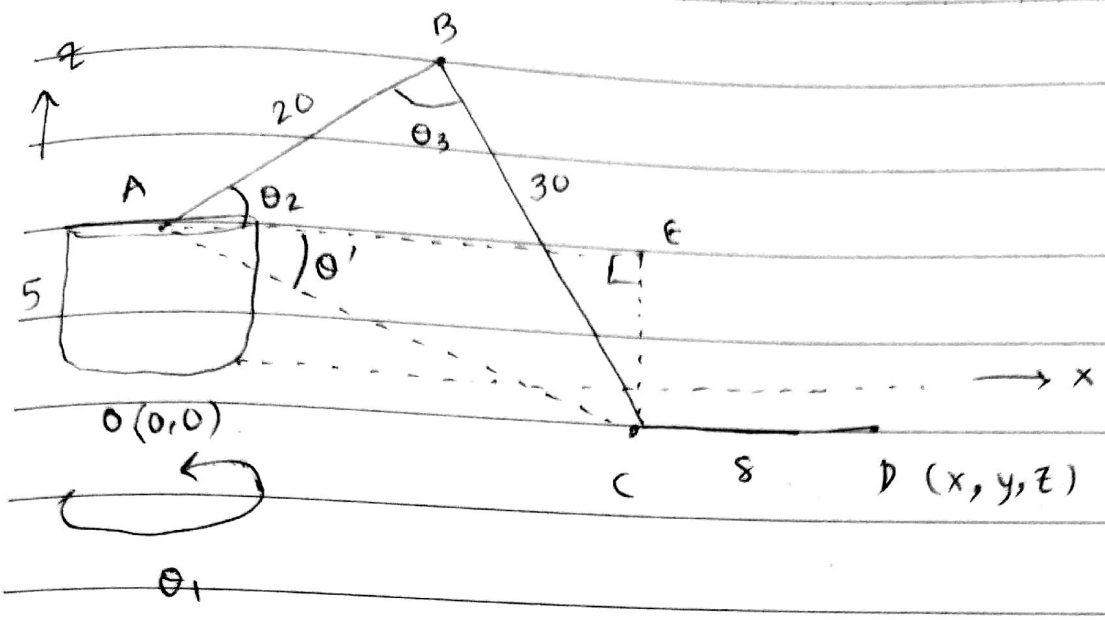
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$\theta_1 = 50^\circ$

$\theta_2 = 40^\circ$

$\theta_3 = 75^\circ$

Side view :

ΔABC,

$$AC^2 = AB^2 + BC^2 - 2.AB.BC \cdot \cos\theta_3$$

$$AC = \sqrt{20^2 + 30^2 - 2.20.30 \cdot \cos(75)} = 31.46$$

Again,

$$\cos\angle BAC = \frac{AB^2 + AC^2 - BC^2}{2.AB.AC}$$

$$\angle BAC = \cos^{-1}\left(\frac{20^2 + 31.46^2 - 30^2}{2 \times 20 \times 31.46}\right) = 67^\circ$$

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$\theta' = \angle BAC - \theta_2$

$= 67 - 40 = 27^\circ$

ΔAEC.

$$\sin\theta' = \frac{CE}{AC}$$

~~$\theta' \neq \sin$~~

$\therefore \quad CE = \sin\theta' \cdot AC$

$= \sin(27) \times 31.46 \quad = 14.28$

$$\cos\theta' = \frac{AE}{AC}$$

$\Rightarrow AE = AC \cos\theta'$

$= 31.46 \times \cos(27) \quad = 28$

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projection,

$d = AE + CD$

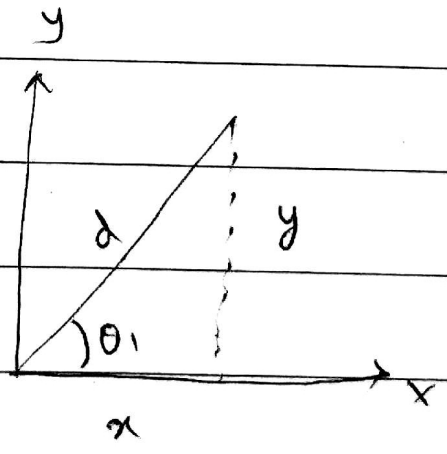
$= 28 + 8$

$= 36$

$z = CE - OA$

$= 14.28 - 5 = 9.28$

Top view :



$\sin\theta_1 = y/d$

$\Rightarrow y = 36 \sin(50)$

$= 27.58$

$\therefore D(23.14, 27.58, -9.28)$

$\cos\theta_1 = x/d$

$x = 36 \cos(50)$

$= 23.14$

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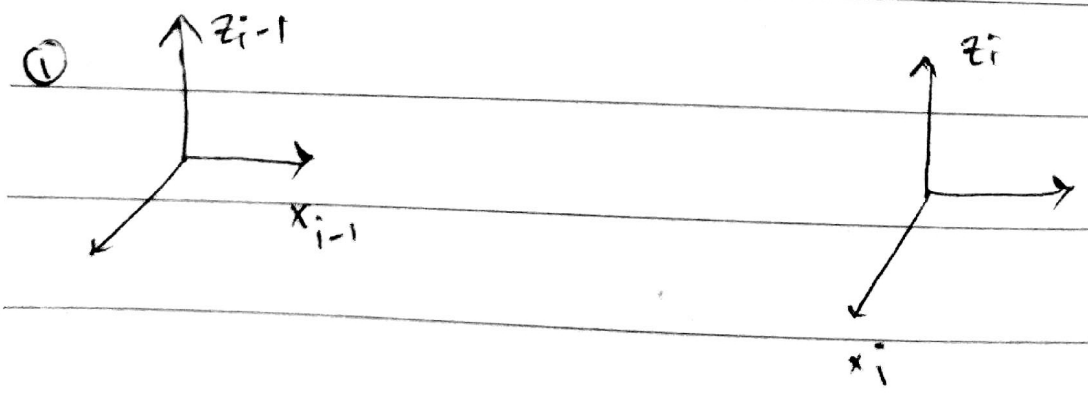
DH Parameter.

* 4 parameters (α, θ, d, a)

angular linear

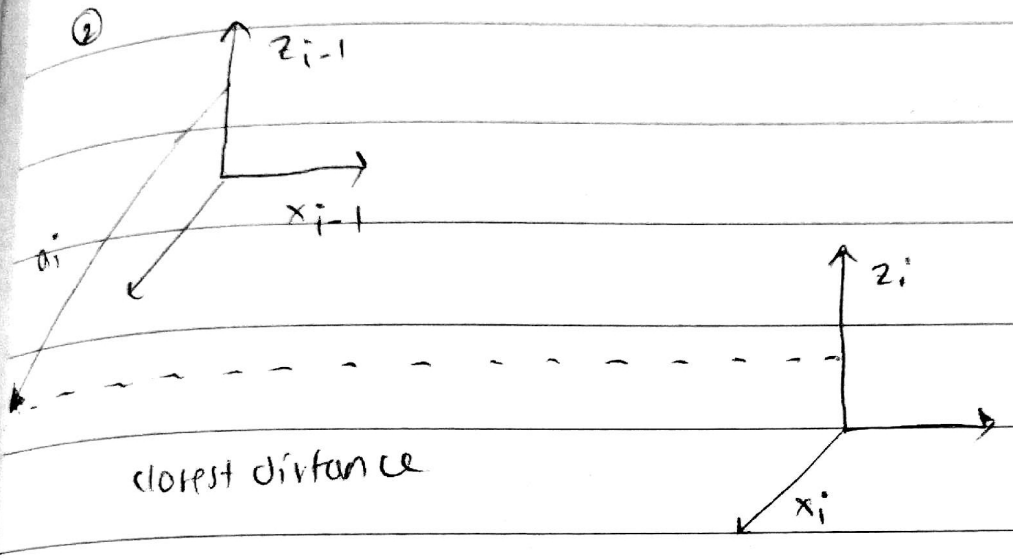
• Link length (a_i)

distance between z_{i-1} and z_i along x_i



Here $a_i = 0$

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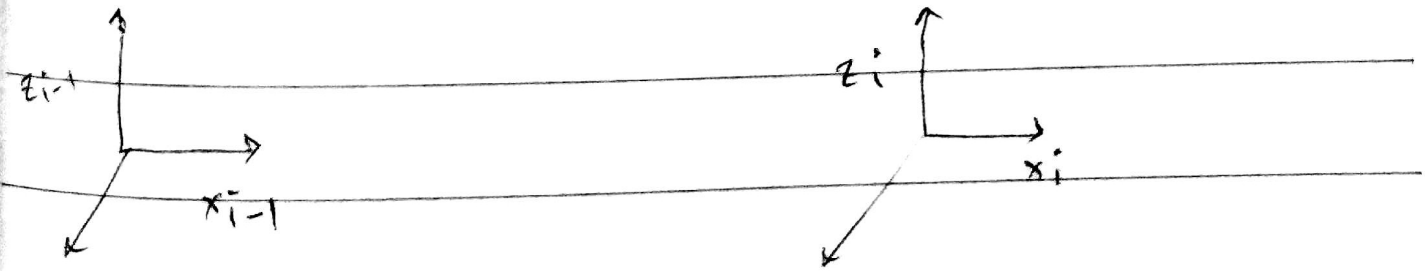
② 

closest distance

between z_{i-1} & z_i

- Joint offset (d_i)

distance between x_{i-1} and x_i along z_{i-1}



$d = 0$

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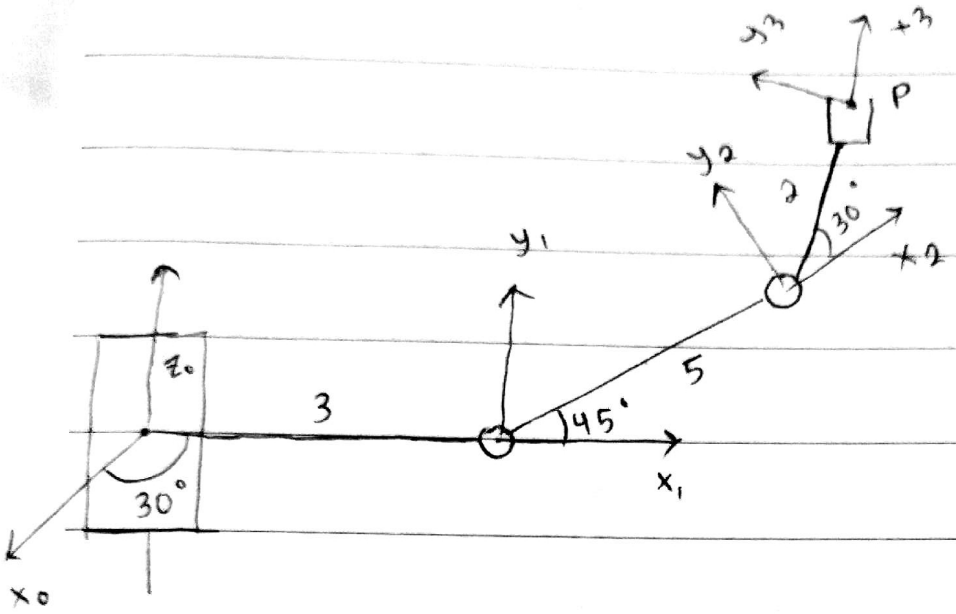
- Link twist (α)

  angle between z_{i-1} and z_i about x_i

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j	a	d	α	θ
1	3	0	90	30°
2	5	0	0	45°
3	2	0	0	30°

$$T = \begin{bmatrix} c\theta & -c\alpha s\theta & s\alpha s\theta & a c\theta \\ s\theta & c\alpha c\theta & -s\alpha c\theta & a s\theta \\ 0 & s\alpha & c\alpha & d \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$$T_1 = \begin{bmatrix} 0.86 & 0 & 0.5 & 2.54 \\ 0.5 & 0 & -0.86 & 1.5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 0.71 & -0.71 & 0 & 3.54 \\ 0.71 & 0.71 & 0 & 3.54 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3 = \begin{bmatrix} 0.87 & -0.5 & 0 & 1.7 \\ 0.5 & 0.87 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T = T_1 \times T_2 \times T_3$$

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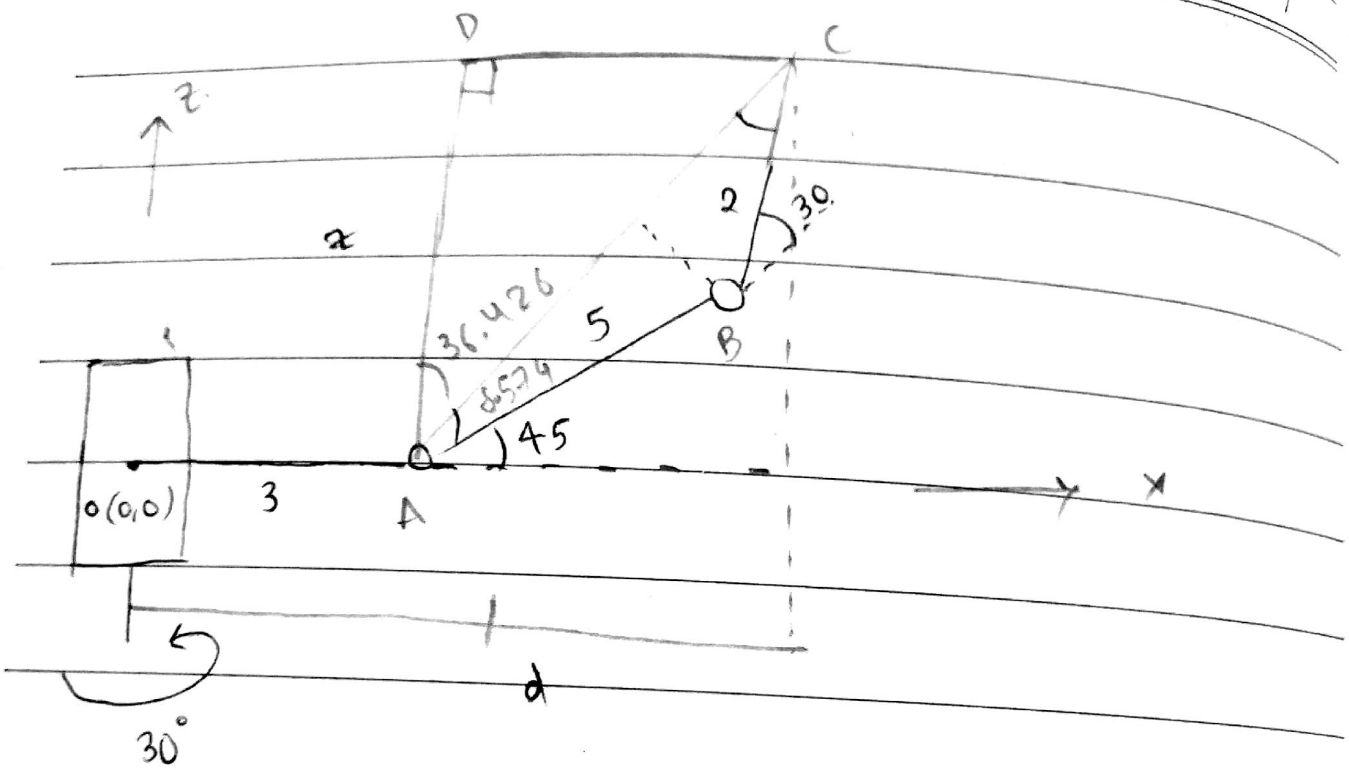
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$$\therefore T = \begin{bmatrix} 0.22 & -0.84 & 0.5 & 6.11 \\ 0.13 & -0.48 & -0.87 & 3.53 \\ 0.97 & 0.26 & 0 & 5.47 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} x \\ y \\ z \\ \end{matrix}$$

side view → Z is value, projection value

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ΔABC,

$\angle ABC = 180 - 30 = 150^\circ$

Again,

$$AC^2 = AB^2 + BC^2 - 2.AB.BC.\cos \angle ABC$$

$$AC = \sqrt{5^2 + 2^2 - 2.5.2\cos(150)}$$

$$\therefore AC = 6.80$$

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Again,

$$\cos \angle CAB = \frac{AB^2 + AC^2 - BC^2}{2 \cdot AB \cdot AC}$$

$$\Rightarrow \angle CAB = \cos^{-1}\left(\frac{5^2 + 6.80^2 - 2^2}{2 \times 5 \times 6.80}\right) = 8.574$$

$$\angle DAC = 90 - 45 - 8.574 = 36.426$$

ΔADC,

$$\cos \angle DAC = \frac{AD}{AC}$$

$$AD = 6.80 \times \cos(36.426) = 5.47 = 2$$

$$\sin \angle DAC = \frac{CD}{AC}$$

$$CD = 6.80 \times \sin(36.426)$$

$$= 4.04$$

projection, d

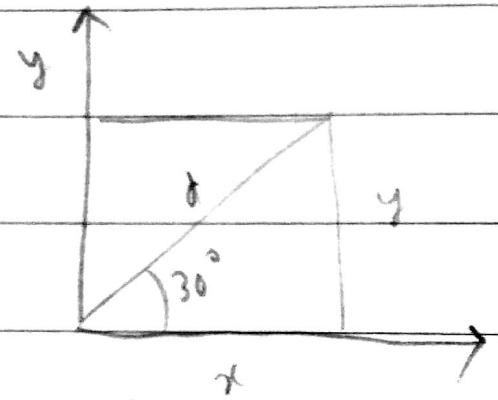
$= 3 + 4.04$

$= 7.04$

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Top view



$\cos 30 = \frac{x}{d}$

$x = 7.04 \times \cos 30 = 6.10$

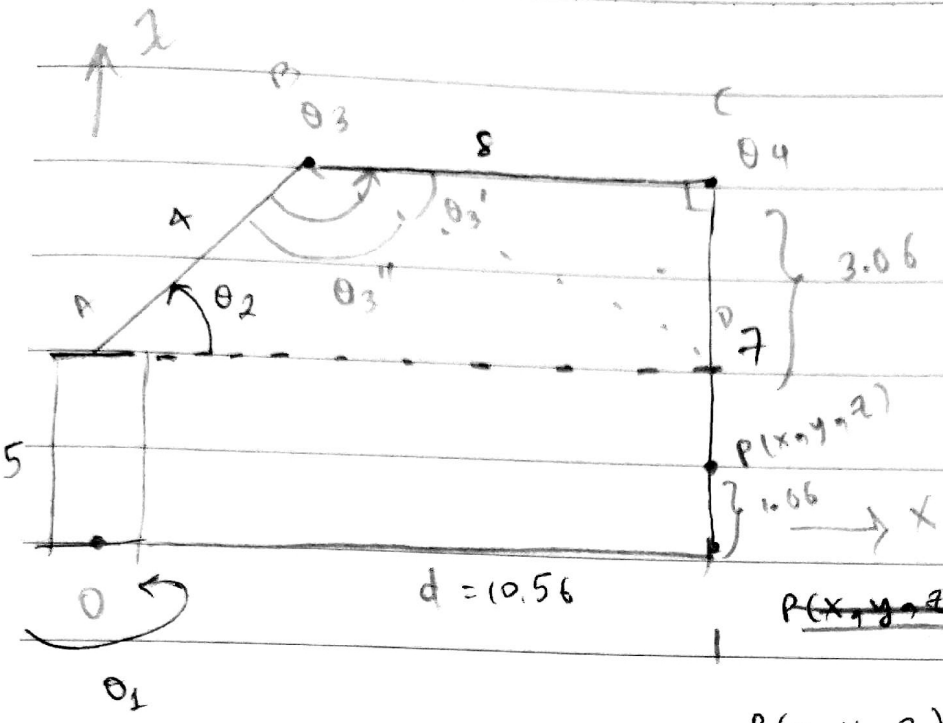
$\sin 30 = y/d$

$y = \sin 30 \times 7.04$

$= 3.52$

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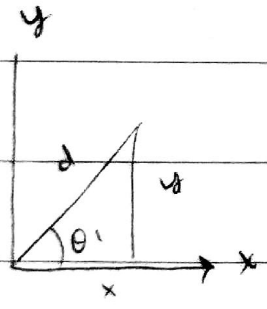
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~~P(x,y,z)~~

$P(x, y, z) = (8.65, 6.06, 1.06)$

Top view:

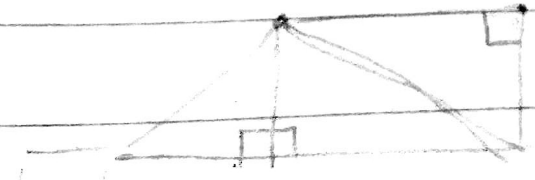


$\tan\theta_1 = y/x$

$\theta_1 = \tan^{-1}(y/x)$

$\tan^{-1}\left(\frac{6.06}{8.65}\right) = 35°$

$d = \sqrt{8.65^2 + 6.06^2} = 10.56$



(another way)

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Side view :

$PD = 5 - 1.06 = 3.94$

$CD = 7 - 3.94 = 3.06$

ΔBCD,

$\tan \theta_3' = CD/BC$

$\theta_3' = \tan^{-1}\left(3.06/8\right) = 20.93^\circ$

$\Rightarrow BD = \sqrt{BC^2 + CD^2} = \sqrt{8^2 + 3.06^2} = 8.56$

ΔABD,

$$\cos \theta_3'' = \frac{AB^2 + BD^2 - AD^2}{2 \cdot AB \cdot BD}$$

$$= \cos^{-1}\left(\frac{4^2 + 8.56^2 - 10.56^2}{2 \cdot 4 \cdot 8.56}\right)$$

$= 108.9^\circ$

$(\because AD = d)$

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$$\cos\theta_2 = \frac{AB^2 + AD^2 - BD^2}{2 \cdot AB \cdot AD}$$

$$\Rightarrow \theta_2 = \cos^{-1}\left(\frac{4^2 + 10.56^2 - 8.56^2}{2.4 \cdot 10.56}\right)$$

$$= 50.05$$

$$\therefore \theta_3 = 108.9 + 20.93$$

$$= 129.83$$